

Regulating Under Uncertainty: the Case of Generative AI with Florence G'sell, Visiting Professor at the Cyber Policy Center

Florence G'sell, holding the digital governance chair at Cen Poe and teaching at the University of Lauron, is the inaugural director of the program on governance of emerging technologies at the Cyber Policy Center. The program is a result of a multi-university initiative examining various issues related to cyber policy.

The talk provides insight into an extensive report on the regulation of generative AI, part of a series including virtual/augmented reality and blockchain. The report's challenge is its timeliness, as the field of generative AI governance is rapidly evolving.

Feedback on the report is sought from the Silicon Valley community and beyond, emphasizing the importance of community input for refining the report. The report addresses policy proposals across three technological domains, highlighting the dynamic nature of policy development in these areas.

Florance's presentation focuses on the complex question of regulating generative AI under conditions of uncertainty. The report began in summer with the EU Parliament's version of the AI Act and has since undergone numerous updates, reflecting the fast-paced changes in AI regulation.

The report discusses generative AI models, including chatbots like ChatGPT, which are based on broader "foundation models." These models are characterized by immense data volumes, fine-tuning capabilities for specific tasks, complexity, and unforeseeable capabilities, which are central to the regulatory challenges.

An example of generative AI's unforeseen capabilities is its ability to solve previously unsolved mathematical problems, indicating the potential and unpredictability of these technologies. However, concerns about risks such as inaccuracy, discrimination, and societal misalignments are also prevalent.

Policymakers face a dilemma: to regulate or not to regulate emerging technologies. One approach is to refrain from regulation, allowing the industry to self-regulate, trusting that industry experts will establish safety standards and best practices.

However, relying solely on industry self-regulation is insufficient due to potential conflicts of interest, as companies prioritize profit and may release untested models. An alternative is a "light touch" approach, where policymakers encourage self-regulation but also leverage existing frameworks to enforce basic standards.

In the U.S., NIST has released AI standards that are being updated for generative AI. While not legally binding, these standards are significant and reflect a more hands-off regulatory approach compared to the EU.

The EU Commission is seeking commitments from major companies to test products before release, be transparent about models, and use techniques like watermarking to indicate synthetic content. These commitments are part of a broader regulatory strategy that includes mandatory frameworks.

The U.S. has issued an executive order mandating the assessment of risks associated with powerful generative AI models. It includes reporting requirements for developers and operators of large-scale computing clusters, introducing measures like red teaming to evaluate AI systems.

The EU's AI Act is poised to have a global impact, as it regulates providers and users of AI systems within and outside the EU, based on where the AI system's output is utilized. This extraterritorial reach means companies worldwide are preparing for compliance.

The AI Act's timeline indicates that banned AI systems will be prohibited in the EU within six months, and provisions for general-purpose AI models will be enforceable in one year. This rapid implementation underscores the EU's commitment to regulating AI technologies.

The AI Act initially focused on use cases, but the release of ChatGPT prompted a shift to include provisions for regulating the technology itself, such as foundation models. Defining these models is challenging, and the EU has adopted the term "general purpose AI models" for this purpose.

The AI Act categorizes AI applications based on risk: unacceptable, high risk, and limited risk. Unacceptable applications, like social scoring, are banned. High-risk applications, such as those in sensitive sectors, face stringent obligations. Limited risk applications require mandatory disclosure, such as informing users when interacting with a chatbot.

The AI Act introduces a new layer of regulation for general-purpose AI systems, particularly those with systemic risks. These systems are defined by their computational resources, and stringent transparency obligations apply to them, including technical documentation and energy consumption disclosures.

The AI Act's approach to open source is ambiguous, with exemptions for certain transparency obligations. However, the definition of open source is narrow, requiring full disclosure of model details, which may limit the scope of what is considered open source under the act.

The AI Act will be enforced at the national level by designated authorities, except in cases involving general-purpose models provided by companies like OpenAI, where the EU Commission is directly competent. This centralization of authority is a notable aspect of the act.

Concurrently with the AI Act, the EU is adopting a directive on product liability for software, addressing the safety expectations for AI models. This directive, while limited to the EU, is an important legal framework complementing the AI Act.

The presentation concludes with a discussion on the importance of academic institutions in explaining complex legal regimes and the potential influence of the EU's regulatory model on global AI governance. The AI Act's potential to set international standards, similar to the GDPR, is highlighted.